

Table 1. Complete list of air handling unit (AHU) domain pairs, suitable for transfer learning (TL) algorithms.

Domains		Corresponding faults		Similarities btw. domains	Discrepancies btw. domains	
1 st	2 nd	1 st domain faults (followed by injection date: mm/dd, if available)	2 nd domain faults (followed by injection date: mm/dd, if available)			
ASHRAE-NIST	RBC-Nesbitt	Summer	CooCoiValStuck_100 (8/31) CooCoiValStuck_65 (9/02)	CooCoiValStuck_100 (08/11) CooCoiValStuck_80 (07/11)	*Same control strategy (RBC). *Both contain real- building datasets.	* Building type and weather condition (ASHRAE-NIST: ERS test facility in Iowa city, Iowa, USA. RBC-Nesbitt: a commercial building in Philadelphia, PA, USA).
			TSup_p2.8 (07/13)	TSup_p4 (09/07)		
			OADamLea_45 (09/05) OADamLea_55 (09/06)	OADamStuck_90 (07/11) OADamStuck_60 (07/18)		
		Winter	OADamLea_62 (02/15)	OADamStuck_90 (12/01) OADamStuck_80 (01/03)		
ASHRAE-NIST	RBC-5wk	Summer	CooCoiValStuck_0 (08/27) CooCoiValStuck_15 (09/01)	CooCoiValStuck_0 CooCoiValStuck_5 CooCoiValStuck_15	*Same control strategy (RBC). *Both represent a medium-sized office building.	*Building type and weather condition (ASHRAE-NIST: ERS test facility in Iowa city, Iowa, USA. RBC-5wk: medium-sized office building in Tuscaloosa, AL, USA).
			CooCoiValStuck_65 (09/02) CooCoiValStuck_100 (08/31)	CooCoiValStuck_100		
			OADamLea_45 (09/05) OADamLea_55 (09/06)	OADamStuck_65 OADamStuck_100		
			SupDucLea_AfterSF (09/07)	SupDucLea_20 SupDucLea_45 SupDucLea_55		
			TSup_P1.7 (07/12) TSup_P2.8 (07/13)	TSup_p2		
ASHRAE-NIST	LBNL-2019-1	Transition	TOut_p3 (5/29)	TOA_p2 (05/16 to 05/21) TOA_p4 (05/23 to 05/28)	*Same control strategy (RBC). *Both represent a medium-sized office building.	* Building type and weather condition (ASHRAE- NIST: ERS test facility in Iowa city, Iowa, USA. LBNL-2019-1: medium-sized office building in Chicago, IL, USA).
			TOut_m3 (05/30)	TOA_m2 (06/06 to 06/11) TOA_m4 (06/13 to 06/18)		

<i>RBC-ASHRAE1312</i>		All faults are available for both real building and its simulation. TL algorithms can be assessed using one domain as the source domain and another as the target domain (simulation vs. real).			*Same building, same control strategy, and same weather condition.	* One is simulated and the other is the real-building dataset.
<i>G36-1wk</i>	<i>LBNL-2019-2</i>	Summer	OADamStuck_0	OADamStuck_min (09/18)	*Same control strategy (G36).	*Building type and weather condition <i>(G36-1wk</i> : medium-sized office building in Chicago, IL, USA. <i>LBNL-2019-2</i> : FLEXLAB test facility in Berkeley, CA, USA). *Data type <i>(G36-1wk</i> : simulated-building data. <i>LBNL-2019-2</i> : real-building data).
			OADamStuck_100	OADamStuck_100 (09/19)		
			CooCoiValStuck_100	CooCoiValStuck_100 (09/22)		
		Winter	HeaCoiValStuck_0 HeaCoiValStuck_5 HeaCoiValStuck_15	HeaCoiValStuck_0 (03/24)		
			HeaCoiValStuck_100	HeaCoiValStuck_100 (03/25)		
			OADamStuck_100	OADamStuck_100 (03/18)		
<i>G36-1wk</i>	<i>G36-HIL</i>	Summer	CooCoiValStuck_0 CooCoiValStuck_5 CooCoiValStuck_15	Phy_CooCoiValStuck_0	*Same control strategy (G36). *Building and weather condition (medium-sized office building in Chicago, IL, USA).	* <i>G36-HIL</i> mimics the real building response to the faults, but <i>G36-1wk</i> is the pure simulation data.
<i>LBNL-2022</i>	<i>G36-Degrad</i>	Summer & Winter	TSup_p2	TSup_p1	*Same control strategy (G36).	*Weather condition (<i>LBNL-2022</i> : Chicago. IL, USA).

		& Transition	TSup_m2	TSup_m1	*Building type (both simulate annual faults of a medium-sized office building).	<i>G36-Degrad</i> : Tuscaloosa, AL, USA). *In <i>G36-Degrad</i> , fault intensities increase with time, but <i>LBNL-2022</i> faults have the same intensity over the entire period.
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